

Exercices 2 : logistic regressions

1. Turnout and rain

You are interested in knowing whether turnout, the participation in elections, is conditional on weather: that is, do people vote less when it rains? To do so, you are provided with a (fictitious) data set on American elections in 2012, 2016, and 2020.

0. Load the following packages :

```
library(ggeffects)
library(ggplot2)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.6
v forcats    1.0.1      v stringr    1.6.0
v lubridate  1.9.4      v tibble     3.3.1
v purrr      1.2.1      v tidyr      1.3.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(rio)
```

1. Using `rio` and the function `import`, import the data from https://github.com/scpo-quantimethods/quantitative_methods_two/raw/main/exercices/exercices2/data/rainfall.dta.
2. Have a look at your data using `glimpse()`. Drop all variables except `voted`, `voted_prev`, `rain_mm`, `age`, `female`, `year`.

3. You want to inspect whether the chances of voting change with the rainfall on the day of the election, `rain_mm`. The variable `vote` tells if people have voted in the elections. What would be your dependent and independent variables ? Are they categorical or continuous ?
4. Count the total number of voters and summarize the `rain_mm` variable. Are there NA's in the data set ?
5. Using `mutate`, `case_when`, prepare the `voted` variable for the logistic regression.
6. Assuming you are estimating a logistic regression, what is the mathematical expression of the probability function as a function of rainfall, denoted $R_m m$? What is the corresponding expression for the logit of that probability? Finally, what is the expression for the odds ratio?
7. Run a logistic regression with one variable to estimate whether the rain changes anything in the probabilities of voting. For a 1-mm increase in the rainfall, how does the likelihood of voting change ? How significant is that result ?
8. Does voting in the previous election change the probabilities of voting in the next one ?
9. Add the variables `voted_prev`, `female`, `age`, `rain_mm`, `as.factor(year)` in the predictors, and compare the three models, using the `stargazer` package;
10. Use the `ggpredict` function from package `ggeffects` to predict the probabilities of voting depending on `rain_mm`. Have a look at your predictions.
11. Use `ggplot` to plot the predicted probabilities with their confidence intervals, with their confidence intervals. Hint : use `+ geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.2)` to display smooth confidence intervals, and `geom_line()` to add a line connecting the dots.

```
# run install.packages() if you haven't the packages already installed
library(ggeffects)
library(ggplot2)
library(tidyverse)
library(rio)
library(stargazer)
```

Please cite as:

Hlavac, Marek (2022). `stargazer`: Well-Formatted Regression and Summary Statistics Tables.

R package version 5.2.3. <https://CRAN.R-project.org/package=stargazer>

```
#1.1
rain_vote <- import("https://github.com/scpo-quantimethods/quantitative_methods_two/raw/main,
```

```
#1.2
```

```
glimpse(rain_vote)
```

```
Rows: 6,000
```

```
Columns: 13
```

```
$ voter_id    <dbl> 1, 1, 1, 2, 2, 2, 3, 3, 3, 4, 4, 4, 5, 5, 5, 6, 6, 6, 7~
$ year        <dbl> 2012, 2016, 2020, 2012, 2016, 2020, 2012, 2016, 2020, 2~
$ county      <chr> "County_05", "County_05", "County_05", "County_39", "Co~
$ age         <dbl> 76, 80, 84, 22, 26, 30, 44, 48, 52, 48, 52, 56, 71, 75,~
$ female      <dbl> 0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 0~
$ party       <chr> "U", "U", "U", "R", "R", "R", "D", "D", "D", "U", "U", ~
$ rain_mm     <dbl> 5.52, 5.66, 3.32, 6.29, 4.55, 4.56, 6.54, 5.89, 7.77, 0~
$ voted       <chr> "voted", "voted", "voted", "did not vote", "did not vot~
$ inperson_eday <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0~
$ early_absentee <dbl> 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
$ mail        <dbl> 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0~
$ rain_mm_prev <dbl> NA, 5.52, 5.66, NA, 6.29, 4.55, NA, 6.54, 5.89, NA, 0.2~
$ voted_prev  <dbl> NA, 1, 1, NA, 0, 0, NA, 0, 0, NA, 0, 0, NA, 1, 1, NA, 0~
```

```
rain_vote <- rain_vote %>% select(voted, voted_prev, rain_mm, age, female, year)
```

```
#1.3
```

```
# DV : voting (binary)
```

```
# IV : raining. Likely exogeneous (uncorrelated with the error)
```

```
#1.4
```

```
table(rain_vote$voted)
```

```
did not vote    voted
         4317         1683
```

```
summary(rain_vote$rain_mm)
```

```
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000    4.540    6.310    6.268    7.960   13.610
```

```
summary(rain_vote)
```

voted	voted_prev	rain_mm	age
Length:6000	Min. :0.0000	Min. : 0.000	Min. :18.00
Class :character	1st Qu.:0.0000	1st Qu.: 4.540	1st Qu.:38.00
Mode :character	Median :0.0000	Median : 6.310	Median :55.00
	Mean :0.2495	Mean : 6.268	Mean :55.15
	3rd Qu.:0.0000	3rd Qu.: 7.960	3rd Qu.:72.00
	Max. :1.0000	Max. :13.610	Max. :92.00
	NA's :2000		

female	year
Min. :0.000	Min. :2012
1st Qu.:0.000	1st Qu.:2012
Median :0.000	Median :2016
Mean :0.493	Mean :2016
3rd Qu.:1.000	3rd Qu.:2020
Max. :1.000	Max. :2020

```
#1.5
# with case_when()
rain_vote <- rain_vote %>% mutate(
  voted = case_when(
    voted == "voted" ~ 1,
    voted == "did not vote" ~ 0
  )
)

#1.6
log_model <- glm(
  data = rain_vote,
  voted ~ rain_mm,
  family = "binomial"
)
exp(log_model$coefficients)-1
```

```
(Intercept)    rain_mm
0.03716785 -0.14940336
```

```
exp(coef(log_model))-1
```

```
(Intercept)    rain_mm
0.03716785 -0.14940336
```

```
#1.7
log_model2 <- glm(
  data = rain_vote,
  voted ~ voted_prev,
  family = "binomial"
)
exp(log_model2$coefficients)-1
```

```
(Intercept)  voted_prev
-0.6342129    1.1209079
```

```
exp(coef(log_model2))-1
```

```
(Intercept)  voted_prev
-0.6342129    1.1209079
```

```
#1.8
log_model3 <- glm(
  data = rain_vote,
  voted ~ voted_prev+female+age+rain_mm+as.factor(year),
  family = "binomial"
)
```

```
#1.8
stargazer(log_model, log_model2, log_model3,
  type = "text",
  dep.var.labels = "Voting Behavior",
  dep.var.caption = c("Voting turnout; 0 = abstention | 1 = voted"),
  star.cutoffs = c(0.05, 0.01, 0.001)
)
```

```
=====
                        Voting turnout; 0 = abstention | 1 = voted
                        -----
                        Voting Behavior
                        (1)      (2)      (3)
                        -----
rain_mm                -0.162***                -0.162***
```

	(0.012)		(0.016)
as.factor(year)2020			-0.100 (0.079)
voted_prev		0.752*** (0.076)	0.194* (0.084)
female			0.348*** (0.074)
age			0.039*** (0.002)
Constant	0.036 (0.075)	-1.006*** (0.041)	-2.407*** (0.169)

Observations	6,000	4,000	4,000
Log Likelihood	-3,463.386	-2,428.188	-2,184.353
Akaike Inf. Crit.	6,930.772	4,860.375	4,380.706

Note: *p<0.05; **p<0.01; ***p<0.001

```
#1.9
predicted_probs <- ggpredict(log_model3, terms = "rain_mm")
```

Data were 'prettified'. Consider using `terms="rain\_mm [all]"` to get smooth plots.

```
predicted_probs
```

```
# Predicted probabilities of voted
```

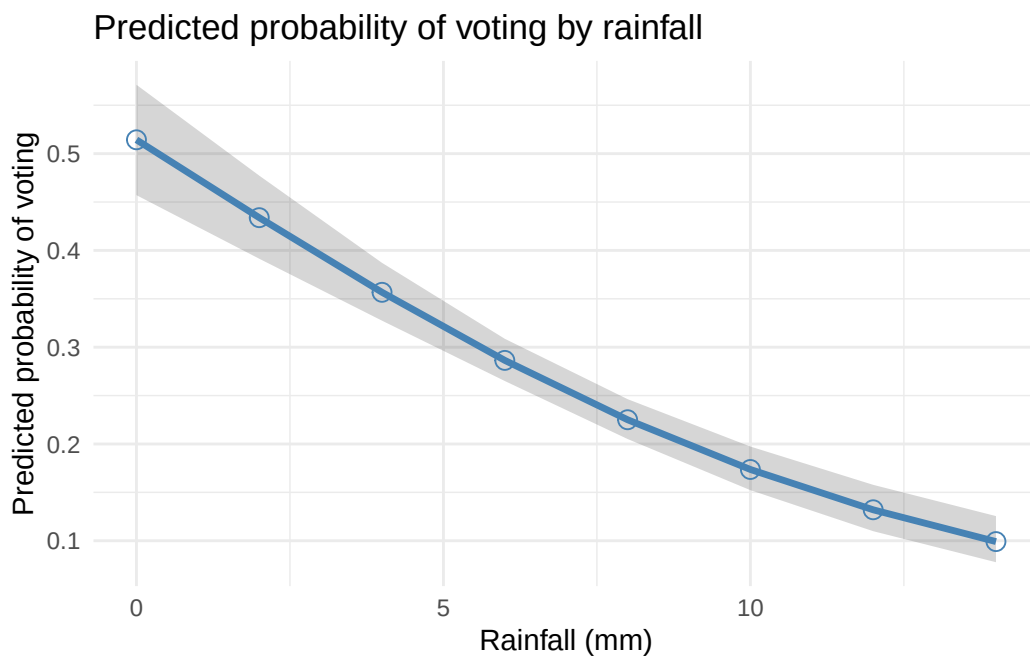
rain_mm	Predicted	95% CI
0	0.51	0.46, 0.57
2	0.43	0.39, 0.48
4	0.36	0.33, 0.39
6	0.29	0.27, 0.31
8	0.23	0.21, 0.25

10		0.17		0.15, 0.20
12		0.13		0.11, 0.16
14		0.10		0.08, 0.13

Adjusted for:

```
* voted_prev = 0.25
*   female = 0.49
*   age = 57.15
*   year = 2016
```

```
#1.10
ggplot(predicted_probs, aes(x = x, y = predicted)) +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.2) + # smooth CI
  geom_line(linewidth = 1.2, color = "steelblue") +
  geom_point(shape = 21, size = 3, color = "steelblue") +
  labs(
    title = "Predicted probability of voting by rainfall",
    x = "Rainfall (mm)",
    y = "Predicted probability of voting"
  ) +
  theme_minimal()
```



2. ESS : voting in Italy

You are provided with a simplified dataset containing ESS data for Italy.

1. Download the dataset and import it into your local environment, using the link https://github.com/scpo-quantimethods/quantitative_methods_two/raw/main/exercices/exercices2/data/ess_it.csv.
2. Inspect the dataset using `glimpse`.
3. Examine the values of the vote variable and modify them using `case_when` if necessary. Prepare the variable for a logistic regression. Review the other variables as well and modify them as needed. You may draw on the theory section for this session for guidance.
4. Run a logistic regression to analyze whether the number of years of education influences the probability of voting.
5. Create a variable `age` and include the variables `polintr`, `trstplt`, `clsprty`, `gn dr`, `age`, `eduyrs` in the logistic regression.
6. Compare the two models using `stargazer`.
7. Plot the predicted probability of voting by years of education.

```
#1.1
ess <- import("https://github.com/scpo-quantimethods/quantitative_methods_two/raw/main/exerc
glimpse(ess)
```

Rows: 2,640

Columns: 8

\$ cntry	<chr>	"IT", "IT", "IT", "IT", "IT", "IT", "IT", "IT", "IT", "IT", "I~
\$ polintr	<int>	3, 3, 3, 3, 4, 3, 3, 3, 3, 3, 4, 2, 1, 2, 4, 3, 3, 2, 3, 3, 2,
\$ trstplt	<int>	1, 0, 0, 1, 3, 2, 0, 4, 2, 1, 0, 7, 2, 2, 2, 0, 5, 3, 5, 2, 7,~
\$ vote	<int>	1, 1, 1, 1, 2, 2, 1, 1, 1, 2, 1, 3, 1, 2, 1, 1, 1, 1, 7, 1, 1,~
\$ clsprty	<int>	2, 2, 1, 2, 2, 2, 1, 2, 2, 2, 2, 1, 1, 2, 2, 2, 2, 1, 2, 1, 2,~
\$ gndr	<int>	2, 2, 1, 1, 1, 2, 1, 1, 1, 2, 2, 2, 1, 1, 1, 2, 1, 1, 2, 2, 1,~
\$ yrbrn	<int>	1943, 1963, 1963, 1984, 1984, 1964, 1960, 1983, 1961, 1993, 19~
\$ eduyrs	<int>	5, 18, 13, 8, 8, 16, 20, 18, 18, 10, 8, 13, 22, 18, 8, 17, 13,~

```
#1.2
table(ess$vote)
```

1	2	3	7	8
1799	567	209	40	25


```

ess <- ess |>
# mutating the dependent variable to get rid of any unnecessary rows
# as we have done above
mutate(
  vote = case_when(
    vote %in% c(3, 7, 8, 9) ~ NA_integer_,
    vote == 2 ~ 0,
    vote == 1 ~ 1,
    TRUE ~ vote
  ),
  # Recode 'polintr' to drop values 7 to 9
  polintr = case_when(
    polintr %in% c(7:9) ~ NA_integer_,
    TRUE ~ polintr
  ),
  # Recode 'clsprty' 1/2 into 0/1 and drop 7, 8
  clsprty = case_when(
    clsprty == 1 ~ 0,
    clsprty == 2 ~ 1,
    clsprty %in% c(7, 8) ~ NA_integer_,
    TRUE ~ clsprty
  ),
  # Recode 'gndr' into 0/1 and handle other cases as NAs
  gndr = case_when(
    gndr == 1 ~ 0,
    gndr == 2 ~ 1,
    TRUE ~ NA_integer_
  ),
  # recode eduyrs
  eduyrs = case_when(
    eduyrs %in% c(1:55) ~ eduyrs,
    TRUE ~ NA_integer_
  )
)
#1.3
table(ess$vote)

```

```

  0    1
567 1799

```

```
#1.3
logit1 <- glm(
  vote ~ eduysrs,
  data = ess %>% mutate(age = 2026-yrbrn),
  family = "binomial")

logit2 <- glm(
  vote ~ polintr + trstplt + clsprty + gndr +age + eduysrs,
  data = ess %>% mutate(age = 2026-yrbrn),
  family = "binomial")

#1.4
# simply use exp() on the coefficients of the logit
exp(coef(logit1))
```

```
(Intercept)      eduysrs
      1.041496      1.098984
```

```
# here would be a second way of doing it
exp(logit1$coefficients)
```

```
(Intercept)      eduysrs
      1.041496      1.098984
```

```
exp(cbind(
  OR = coef(logit1),
  confint(logit1)))
```

Waiting for profiling to be done...

```
              OR      2.5 %   97.5 %
(Intercept) 1.041496 0.7779327 1.394698
eduysrs      1.098984 1.0734494 1.125548
```

```
#1.5
stargazer(logit1, logit2,
  type = "text",
  dep.var.labels = "Voting Behavior",
  dep.var.caption = c("Voting turnout; 0 = abstention | 1 = voted"),
  star.cutoffs = c(0.05, 0.01, 0.001))
```

```

=====
                        Voting turnout; 0 = abstention | 1 = voted
                        -----
                                Voting Behavior
                                (1)                (2)
                        -----
polintr                                -0.704***
                                         (0.080)

trstplt                                -0.006
                                         (0.007)

clsprty                                -1.507***
                                         (0.177)

gndr                                   0.116
                                         (0.112)

age                                    0.00001
                                         (0.0001)

eduyrs                                0.094***
                                0.012)    (0.014)
                                         (0.014)

Constant                                3.820***
                                0.149)    (0.348)
                                         (0.348)

                        -----
Observations                            2,284                2,175
Log Likelihood                         -1,214.665            -1,032.654
Akaike Inf. Crit.                      2,433.330            2,079.309
=====
Note:                                *p<0.05; **p<0.01; ***p<0.001

```

```

#1.6
predicted_probs <- ggpredict(logit2, terms = "eduyrs")

```

Data were 'prettified'. Consider using `terms="eduyrs [all]"` to get smooth plots.

```
predicted_probs
```

```
# Predicted probabilities of vote
```

```
eduyrs | Predicted |      95% CI
-----|-----|
    2 |      0.69 | 0.63, 0.75
    6 |      0.74 | 0.70, 0.78
    8 |      0.76 | 0.73, 0.79
   12 |      0.80 | 0.78, 0.82
   16 |      0.83 | 0.81, 0.85
   18 |      0.85 | 0.82, 0.87
   22 |      0.87 | 0.84, 0.90
   28 |      0.91 | 0.86, 0.94
```

```
Adjusted for:
```

```
* polintr = 3.00
* trstplt = 3.00
* clsprty = 0.70
*   gndr = 0.53
*   age = -15.62
```

Not all rows are shown in the output. Use ``print(..., n = Inf)`` to show all rows.

```
ggplot(predicted_probs, aes(x = x, y = predicted)) +
  geom_line(color = "blue", size = 1) +
  geom_point() +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high), fill = "grey", alpha = 0.3) +
  labs(
    title = "Predicted Probability of Voting by education years",
    x = "Education years",
    y = "Predicted Probability of Voting"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 14, face = "bold", hjust = 0.5),
    axis.title = element_text(size = 12),
    axis.text = element_text(size = 10)
  )
```

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.

